

Corollary: Let T be a normal linear transformation on V . Then eigenvectors of T associated with distinct eigenvalues are orthogonal.

Proof: Suppose $T\vec{v}_1 = \lambda_1\vec{v}_1$ and $T\vec{v}_2 = \lambda_2\vec{v}_2$ with $\vec{v}_1 \neq \vec{0}$ and $\vec{v}_2 \neq \vec{0}$, and $\lambda_1 \neq \lambda_2$.

$$\text{Then } (T\vec{v}_1, \vec{v}_2) = (\lambda_1\vec{v}_1, \vec{v}_2) = \lambda_1(\vec{v}_1, \vec{v}_2)$$

$$(T\vec{v}_1, \vec{v}_2) = (\vec{v}_1, T^*\vec{v}_2) = (\vec{v}_1, \bar{\lambda}_2\vec{v}_2) = \bar{\lambda}_2(\vec{v}_1, \vec{v}_2)$$

$$\Rightarrow \lambda_1(\vec{v}_1, \vec{v}_2) = \bar{\lambda}_2(\vec{v}_1, \vec{v}_2)$$

$$\Rightarrow (\lambda_1 - \bar{\lambda}_2)(\vec{v}_1, \vec{v}_2) = 0$$

Since $\lambda_1 \neq \lambda_2$, $\lambda_1 - \bar{\lambda}_2 \neq 0$.

$(\vec{v}_1, \vec{v}_2) = 0 \Leftrightarrow \vec{v}_1$ and $\vec{v}_2$ are orthogonal.

Corollary. Let $A \in \mathbb{C}^{n \times n}$ be a normal matrix. Then eigenvectors of A associated with distinct eigenvalues are orthogonal.

Proof: Homework.

Schur's Lemma. Let $A \in \mathbb{C}^{n \times n}$. Then there exists a unitary matrix

$P \in \mathbb{C}^{n \times n}$ and an upper triangular matrix U such that

$$A = PUP^*.$$

Remarks: Similar proof as the other version of Schur lemma, where P was only invertible.

One notices that in that proof by induction, one may assume P_1 to be unitary in the base case by choosing

$\vec{x}_1$ to be a unit vector, and extending $\vec{x}_1$ to an orthonormal basis such that P_1 is unitary.

② Then in the inductive hypothesis, one may assume Q to be unitary as well. $\square$

Another proof: By previous Schur's Factorization, $\forall A \in \mathbb{C}^{n \times n}$, there exists an invertible matrix P and an upper triangular matrix U such that $A = PUP^{-1}$.

Let $P = QR$ be the QR factorization. Then Q is orthogonal and R is upper triangular

$$\begin{aligned} \Rightarrow A &= QRUR^{-1}Q^{-1} \\ &= Q(RUR^{-1})Q^{-1} \end{aligned}$$

Note that ① the inverse of an upper triangular matrix is still an upper triangular matrix,

② the product of upper triangular matrices is still upper triangular

$\Rightarrow RUR^{-1}$ is upper triangular.

$\Rightarrow A = Q(RUR^{-1})Q^{-1}$ with Q orthogonal and RUR^{-1} upper triangular.

Schur's Lemma: Let T be a linear transformation on a finite dimensional inner product space V . Then there exists an orthonormal basis B such that $[T]_B$ is an upper triangular matrix.

Spectral Thm for normal matrices:

$A \in \mathbb{C}^{n \times n}$ is a normal matrix. if and only if there exists a unitary matrix P and a diagonal matrix D such that $A = PDP^*$.

Proof. " $\Leftarrow$ " If $A = PDP^*$ for some P unitary and D diagonal

$$\begin{aligned}\text{Then } A^*A &= (PDP^*)^* PDP^* \\ &= P D^* P^* P D P^* = P D^* D P^*\end{aligned}$$

$$AA^* = P D P^* (P D P^*)^* = P D P^* P D^* P^* = P D D^* P^*$$

$$\text{Since } D \text{ is diagonal } D^* D = D D^*$$

$$\Rightarrow AA^* = A^*A$$

i.e. A is normal

By Schur's Lemma, there exists a unitary matrix P and an upper triangular matrix U such that $A = PUP^*$.

Claim: U is actually a diagonal matrix.

Proof: Denote $P = (\vec{p}_1 \ \vec{p}_2 \ \dots \ \vec{p}_n)$ and $U = \begin{pmatrix} \lambda_1 & u_{12} & \dots & u_{1n} \\ & \lambda_2 & \dots & u_{2n} \\ & & \ddots & \vdots \\ & & & \lambda_n \end{pmatrix}$

On one hand, we have

$$A = PUP^* \Leftrightarrow AP = PU$$

$$A(\vec{p}_1 \ \vec{p}_2 \ \dots \ \vec{p}_n) = (\vec{p}_1 \ \vec{p}_2 \ \dots \ \vec{p}_n) \begin{pmatrix} \lambda_1 & u_{12} & \dots & u_{1n} \\ & \lambda_2 & \dots & u_{2n} \\ & & \ddots & \vdots \\ & & & \lambda_n \end{pmatrix}$$

$$(A\vec{p}_1 \ A\vec{p}_2 \ \dots \ A\vec{p}_n) = \begin{pmatrix} \lambda_1 \vec{p}_1 & u_{12}\vec{p}_1 + \lambda_2 \vec{p}_2 & \dots & u_{1n}\vec{p}_1 + \dots + \lambda_n \vec{p}_n \end{pmatrix}$$

$$\Leftrightarrow A\vec{p}_1 = \lambda_1 \vec{p}_1 \quad \textcircled{1}$$

$$A\vec{p}_k = \lambda_k \vec{p}_k + \sum_{i=1}^{k-1} u_{ik} \vec{p}_i \quad \text{for all } k=2, \dots, n$$

On the other hand: $A = PUP^* \Leftrightarrow A^* = (P^*)^* U^* P^* = P U^* P^*$

$$\Leftrightarrow A^* P = P U^*$$

$$\Leftrightarrow (A^* \vec{p}_1 \ A^* \vec{p}_2 \ \dots \ A^* \vec{p}_n) = (\vec{p}_1 \ \dots \ \vec{p}_n) \begin{pmatrix} \bar{\lambda}_1 & & & \\ \bar{u}_{12} & \bar{\lambda}_2 & & \\ \vdots & \vdots & \ddots & \\ \bar{u}_{1n} & \bar{u}_{2n} & \dots & \bar{\lambda}_n \end{pmatrix}$$

$$\Leftrightarrow A^* \vec{p}_1 = \bar{\lambda}_1 \vec{p}_1 + \sum_{i=2}^n \bar{u}_{1i} \vec{p}_i \quad \textcircled{2}$$

$$A^* \vec{p}_k = \bar{\lambda}_k \vec{p}_k + \sum_{i=k+1}^n \bar{u}_{ki} \vec{p}_i$$

$$A^* \vec{p}_n = \bar{\lambda}_n \vec{p}_n$$

By ① $A\vec{p}_1 = u_{11} \vec{p}_1 \Rightarrow (\lambda_1, \vec{p}_1)$ is an eigenvalue/eigenvector pair of A

By the lemma $(\bar{\lambda}_1, \vec{p}_1)$ is an eigenvalue/eigenvector pair of A^* , that is $A^* \vec{p}_1 = \bar{\lambda}_1 \vec{p}_1$.

By ②: $A^* \vec{p}_1 = \bar{\lambda}_1 \vec{p}_1 + \bar{u}_{12} \vec{p}_2 + \dots + \bar{u}_{1n} \vec{p}_n$

$$\Rightarrow \bar{\lambda}_1 \vec{p}_1 = \bar{\lambda}_1 \vec{p}_1 + \bar{u}_{12} \vec{p}_2 + \dots + \bar{u}_{1n} \vec{p}_n$$

$$\Leftrightarrow \bar{u}_{12} \vec{p}_2 + \dots + \bar{u}_{1n} \vec{p}_n = \vec{0}$$

As $\{\vec{p}_2, \dots, \vec{p}_n\}$ is linearly independent, $\bar{u}_{12} = \dots = \bar{u}_{1n} = 0$

$$\Rightarrow u_{12} = \dots = u_{1n} = 0.$$

$$U = \begin{pmatrix} \lambda_1 & u_{11} & u_{12} & u_{13} & \dots & u_{1n} \\ & \lambda_2 & u_{23} & u_{24} & \dots & u_{2n} \\ & & \lambda_3 & u_{34} & \dots & u_{3n} \\ & & & \ddots & \ddots & \ddots \\ & & & & \lambda_n & \end{pmatrix}$$

↓ step 1
step 2
step 3

$$\text{In } AP=PU: \quad A\vec{p}_2 = u_{12}\vec{p}_1 + \lambda_2\vec{p}_2 = \lambda_2\vec{p}_2 \quad \text{as } u_{12}=0.$$

$$\Rightarrow A^*\vec{p}_2 = \bar{\lambda}_2\vec{p}_2 \quad \text{by the lemma.}$$

$$\text{In } A^*P=PU^*: \quad A^*\vec{p}_2 = \bar{\lambda}_2\vec{p}_2 + \bar{u}_{23}\vec{p}_3 + \dots + \bar{u}_{2n}\vec{p}_n$$

$$\Rightarrow \lambda_2\vec{p}_2 = \bar{\lambda}_2\vec{p}_2 + \bar{u}_{23}\vec{p}_3 + \dots + \bar{u}_{2n}\vec{p}_n$$

$$\Rightarrow \bar{u}_{23} = \dots = \bar{u}_{2n} = 0 \quad (\Leftrightarrow) \quad u_{23} = \dots = u_{2n} = 0.$$

$$\text{In } AP=PU: \quad A\vec{p}_3 = u_{13}\vec{p}_1 + u_{23}\vec{p}_2 + \lambda_3\vec{p}_3 = \lambda_3\vec{p}_3 \quad \text{as } u_{13}=u_{23}=0$$

$$\Rightarrow A^*\vec{p}_3 = \bar{\lambda}_3\vec{p}_3 \quad \text{by the lemma.}$$

$$\text{In } A^*P=PU^*: \quad A^*\vec{p}_3 = \bar{\lambda}_3\vec{p}_3 + \bar{u}_{34}\vec{p}_4 + \dots + \bar{u}_{3n}\vec{p}_n$$

$$= \bar{\lambda}_3\vec{p}_3$$

$$\Rightarrow u_{34} = \dots = u_{3n} = 0$$

Continues this way, each step we will get $u_{k,k+1} = \dots = u_{k,n} = 0$, for all $k=1, \dots, n-1$.



^ Spectral Thm for normal linear transformation.

Let V be a finite dimensional inner product space. Let $T: V \rightarrow V$ be a linear transformation. The T is normal if and only if there exists an orthonormal basis B of V such that $[T]_B$ is a diagonal matrix.

Proof: " $\Leftarrow$ " Suppose there exists an orthonormal basis B such that $[T]_B = D$ is diagonal

$$[T^*T]_B = [T^*]_B [T]_B = ([T]_B)^* ([T]_B) = D^* D$$

$$\text{and } [TT^*]_B = [T]_B [T^*]_B = [T]_B ([T]_B)^* = D D^*$$

Since D is diagonal, then $D^*D = DD^*$

$$\Rightarrow [T^*T]_{\mathcal{B}} = [TT^*]_{\mathcal{B}}$$

$$\Rightarrow T^*T = TT^*.$$

" $\Leftarrow$ " Let $\mathcal{B}'$ be an orthonormal basis of V . Denote $[T]_{\mathcal{B}'} = A$

Then since $T^*T = TT^*$

$$\Rightarrow [T^*T]_{\mathcal{B}'} = [T]_{\mathcal{B}'}^* [T]_{\mathcal{B}'} = [TT^*]_{\mathcal{B}'} = [T]_{\mathcal{B}'} ([T]_{\mathcal{B}'})^*$$

$$\Leftrightarrow A^*A = AA^*$$

$\Leftrightarrow A$ is normal

Therefore there exists a unitary matrix P and a diagonal matrix D

such that $A = PDP^*$

(Recall: Let $\mathcal{B}' = \{\vec{b}'_1, \dots, \vec{b}'_n\}$ and $\mathcal{B} = \{\vec{b}_1, \dots, \vec{b}_n\}$ are both bases of V .)

Assume $[\vec{b}_i]_{\mathcal{B}'} = \vec{p}_i$ and take $P = (\vec{p}_1 \dots \vec{p}_n)$

$$\text{Then } [T]_{\mathcal{B}} = P^* [T]_{\mathcal{B}'} P$$

Here take $\{\vec{b}_1, \dots, \vec{b}_n\}$ such that $[\vec{b}_i]_{\mathcal{B}'} = \vec{p}_i = \begin{pmatrix} p_{1i} \\ \vdots \\ p_{ni} \end{pmatrix} \Leftrightarrow \vec{b}_i = \sum_{k=1}^n p_{ki} \vec{b}'_k$

$$\Rightarrow (\vec{b}_i, \vec{b}_j) = \left(\sum_{k=1}^n p_{ki} \vec{b}'_k, \sum_{m=1}^n p_{mj} \vec{b}'_m \right)$$

$\downarrow \{\vec{b}'_1, \dots, \vec{b}'_n\} \text{ is orthonormal}$

$$= \sum_{k=1}^n (p_{ki} \vec{b}'_k, p_{kj} \vec{b}'_k) = \sum_{k=1}^n p_{ki} \bar{p}_{kj} (\vec{b}'_k, \vec{b}'_k)$$

$$= \sum_{k=1}^n p_{ki} \bar{p}_{kj} = (\vec{p}_i, \vec{p}_j) = \begin{cases} 0, & i \neq j \\ 1, & i = j. \end{cases}$$

$\Rightarrow \mathcal{B} = \{\vec{b}_1, \dots, \vec{b}_n\}$ is orthonormal $\uparrow$ $\{\vec{p}_1, \dots, \vec{p}_n\}$ is orthonormal

$$\Rightarrow [T]_{\mathcal{B}} = P^* [T]_{\mathcal{B}'} P = P^* A P = D \rightarrow \text{diagonal.}$$

$\square$

Spectral Theorem for Hermitian Matrices.

A matrix $A \in \mathbb{C}^{n \times n}$ is Hermitian if and only if there exists $P \in \mathbb{C}^{n \times n}$ unitary
 $D \in \mathbb{R}^{n \times n}$ diagonal such that $A = PDP^*$.

Proof: " $\Leftarrow$ ": If $A = PDP^*$ for some P unitary and $D \in \mathbb{R}^{n \times n}$ diagonal, then

$$A^* = (PDP^*)^* = (P^*)^* D^* P^* = PD^* P^* = PDP^* = A$$

$\Rightarrow A$ is Hermitian.

" $\Rightarrow$ ": If A is Hermitian. Then by spectral theorem, there exists P unitary and

D diagonal such that $A = PDP^*$.

$$\text{Since } A^* = (PDP^*)^* = PD^* P^* = A = PDP^* \Rightarrow D = D^*$$

$$\Rightarrow D \in \mathbb{R}^{n \times n}.$$

Spectral Theorem for symmetric matrices.

$A \in \mathbb{R}^{n \times n}$ is symmetric if and only if there exists an orthogonal matrix P and a diagonal
matrix $D \in \mathbb{R}^{n \times n}$ such that $A = PDP^T$

// end of Feb 26.